

10

Contract Theory

In this chapter we explore contexts in which agents interact under asymmetric information. We focus on two main contexts: (1) settings in which an agent does not observe the actions of the other individual(s), and (2) settings in which an agent does not observe the payoff of the other individual(s). The first type of information problem is referred to as “hidden action,” since the uninformed individual (e.g., a manager in a firm) cannot observe the action that another individual exerts (e.g., the effort of employees in the firm) even if the manager is perfectly informed about the type of worker he deals with (e.g., the worker’s ability or productivity). In this context the worker might have incentives to slack from exerting a costly effort, thus giving rise to moral hazard problems. (That is why these settings are also known as “hidden action” or “moral hazard” problems). In this context the manager can offer contracts that provide incentives to the worker to work hard, such as paying a higher salary (bonus) if the worker’s output is high but a low salary otherwise. Providing incentives to work hard is thus costly for the manager, giving rise to a trade-off: higher salaries induce high effort (increasing the firm’s output) but are more expensive to implement than a low effort level. As a consequence the manager only induces a high effort if the firm’s expected profits are higher than those of inducing a low effort. Another context in which moral hazard problems abound is insurance markets, since an insurance company does not observe how careful the insured party is (e.g., the driver’s carefulness is a hidden action for the company) but seeks to give incentives to its clients so that they are as careful as possible.¹

The second type of information problem is referred to as “hidden information,” since in this case one party does not observe another party’s payoff function. Continuing with the example above, the manager might not observe the worker’s ability.

¹ In recent years insurance companies, such as Progressive in the United States, have started to partly observe their clients’ driving patterns with electronic devices such as Snapshot by Progressive. This information is, however, relatively limited as it only records how many miles the client drives each day, when he slams the brakes in sudden stops, and how often he drives between midnight and 4 am, but does not record the client’s driving speed or sudden accelerations.

This information setting is also labeled as “adverse selection,” since the manager could err in its selection of candidates for a job if he does not observe their ability. Other examples include contracting problems between a bank lending money to a client (e.g., entrepreneur) without observing the client’s probability of success in his project. In this context we study menus of contracts that induce each type of worker to select only one contract, so that by their choice they reveal their unobserved type (hidden characteristic) to the manager. For instance, the firm could offer two contracts, one with a high salary but involving a high effort level and another with low salary and low effort, achieving self-selection by each type of worker, whereby the worker with high (low) productivity would select the first (second) type of contract.

In section 10.1, we discuss moral hazard problems, and how their results are affected by the principal’s and agent’s risk attitudes. We analyze discrete and continuous choices, and allow for one or multiple signals of effort. In section 10.2, we examine the “lemons” problem, and show the market inefficiencies that can arise when buyers and sellers have access to different amounts of information. Last, in section 10.3, we explore the adverse selection problem when a principal contracts an agent whose productivity the former cannot observe.

10.1 Moral Hazard

In this section we consider optimal contracts that a principal (e.g., firm) offers to an agent (e.g., worker) in order to induce him to exert a profit-maximizing effort level. We first discuss the basic setup of the model, then we analyze the optimal contract when the principal perfectly observes the effort that the agent exerts, and finally examine how such a contract varies when the principal cannot observe the agent’s effort.

10.1.1 The Model

Consider a principal with benefit function $B(\pi - w)$, where π represents the profit that arises from the agent’s effort (e.g., output or sales), and w denotes the salary that the principal pays to the agent. The benefit function is assumed to be increasing and concave, whereby $B' \geq 0$ and $B'' \leq 0$. The agent’s utility function is, for simplicity, assumed to be quasi-linear,

$$U(w, e) = u(w) - g(e),$$

where $u(w)$ represents the utility that the agent derives from the salary he receives, which is increasing and concave, $u' > 0$ and $u'' \leq 0$; while $g(e)$ denotes his disutility from effort, which is increasing and convex, $g' > 0$ and $g'' \geq 0$.

Importantly, the effort level that the agent exerts affects the probability that a certain profit occurs. In particular, for a given effort e (e.g., 10 hours of work) the conditional probability that a profit of exactly $\pi = \pi_i$ happens (e.g., \$1 million), is

$$f(\pi_i|e) = \text{Prob}\{\pi = \pi_i|e\} \geq 0 \quad \text{for every profit level } \pi_i$$

where $i = \{1, 2, \dots, N\}$ denotes the profits that can emerge for a given effort e . Hence the property above entails that a high profit could arise even if the worker slacks (exerts low effort). In other words, a given profit level $\pi = \pi_i$ can arise from every effort level. We next explore the principal's optimal contract if it could perfectly observe the agent's effort, e .

10.1.2 Symmetric Information

When the principal can observe the agent's effort level, e , its maximization problem becomes

$$\begin{aligned} & \max_{\{e, w(\pi_i)\}_{i=1}^N} \sum_{i=1}^N f(\pi_i|e) \cdot B(\pi_i - w(\pi_i)) \\ & \text{subject to } \sum_{i=1}^N f(\pi_i|e) [u(w(\pi_i)) - g(e)] \geq \bar{u}. \end{aligned} \quad (\text{PC})$$

That is, the principal seeks to maximize expected profits, subject to the agent participating in the contract. The constraint of the program above, as it guarantees the agent's voluntary participation, is often referred to as the participation constraint (PC) or the "individual rationality" condition. In this context, such a constraint must be binding (holding with equality). Otherwise, the principal could lower the salary he offers to the agent and still guarantee that such constraint holds. Regarding the choice variables in the program, note that the principal chooses a salary for each realized profit π_i , $w(\pi_i)$ for all $i = \{1, 2, \dots, N\}$ and an optimal effort level that such list of "contingent salaries" induces the agent to select. Hence we can use the following Lagrangian to solve the above maximization problem

$$\mathcal{L} = \sum_{i=1}^N f(\pi_i|e) \cdot B(\pi_i - w(\pi_i)) + \lambda \left[\sum_{i=1}^N f(\pi_i|e) [u(w(\pi_i)) - g(e)] - \bar{u} \right].$$

In our subsequent discussion, we focus on profit-maximizing salaries for each profit level, that is, for $\{w(\pi_i)\}_{i=1}^N$, but do not directly analyze the other choice variable (which effort level should the principal induce to maximize its expected profits). Once we find optimal salaries, $\{w(\pi_i)\}_{i=1}^N$, we can plug them into the principal's

objective function and solve for the effort level that maximizes the principal's profit (in a two-step approach).

We can now take FOC with respect to w and obtain, for every profit level π_i ,

$$f(\pi_i|e) \cdot B'(\pi_i - w(\pi_i)) \cdot (-1) + \lambda f(\pi_i|e) u'(w(\pi_i)) = 0,$$

where B' and u' denote the derivative of functions $B(\cdot)$ and $u(\cdot)$ with respect to w . Upon rearranging, we have

$$\lambda u'(w(\pi_i)) = B'(\pi_i - w(\pi_i)),$$

and solving for λ yields

$$\lambda = \frac{B'(\pi_i - w(\pi_i))}{u'(w(\pi_i))}, \quad (10.1)$$

which is positive, given that the principal's benefit function is increasing by definition, $B' > 0$, and so is the agent's utility function, $u' > 0$. Since this entails that $\lambda > 0$, the agent's participation constraint must bind (i.e., hold with equality). That is, in expectation, the agent obtains his reservation utility, $\bar{u}$,

$$\sum_{i=1}^N f(\pi_i|e) u(w(\pi_i)) - g(e) = \bar{u}.$$

Example 10.1: Finding optimal wages under moral hazard in symmetric information settings Consider a risk-neutral principal hiring a risk-averse agent with utility function $u(w) = \sqrt{w}$, disutility of effort $g(e) = e$, and reservation utility $\bar{u} = 9$. There are two effort levels $e_H = 5$ and $e_L = 0$. When $e_H = 5$, the principal's sales are \$0 with probability 0.1, \$100 with probability 0.3, and \$400 with probability 0.6. When $e_L = 0$, the principal's sales are \$0 with probability 0.6, \$100 with probability 0.3, and \$400 with probability 0.1. In the case of $e_H = 5$, the expected profit is \$270, while in the case of $e_L = 0$, the expected profit is \$70. When effort is observable, the principal can induce an effort $e_H = 5$ by paying a wage w_e^* that solves

$$u(w_{e_H}^*) = \bar{u} + g(e),$$

which in this setting entails

$$\sqrt{w_{e_H}^*} = 9 + 5 \Leftrightarrow w_{e_H}^* = 14^2 = 196.$$

Similarly the principal can induce a low effort $e_L = 0$ by offering a wage

$$\sqrt{w_{e_L}^*} = 9 + 0 \Leftrightarrow w_{e_L}^* = 9^2 = 81.$$

Given these salaries, the profits that the principal obtains are

$$\$270 - \$196 = \$74 \quad \text{from } e_H = 5,$$

$$\$70 - \$81 = -\$11 \quad \text{from } e_L = 0,$$

and thus he prefers to induce e_H when effort is observable. ■

10.1.3 Risk Attitudes in a Moral Hazard Setting

Continuing with the moral hazard setting under symmetric information, let us now consider the role of risk aversion. We start our analysis with the common case in which the principal is assumed to be risk neutral while the agent is risk averse.

Case 1: Principal is risk neutral but agent is risk averse In this case the principal's benefit function is $B(\pi_i - w(\pi_i)) = \pi_i - w(\pi_i)$, implying that the derivative of $B(\cdot)$ with respect to w collapses to $B'(\pi_i - w(\pi_i)) = -1$. Hence the FOC in expression (1) becomes

$$\lambda = \frac{1}{u'(w(\pi_i))} \quad \text{for all profit levels } \pi_i. \quad (10.2)$$

Figure 10.1 depicts this result. (Recall that $u'' \leq 0$ which entails that u' is weakly decreasing in w , and thus its inverse $1/u'$ is weakly increasing in w).

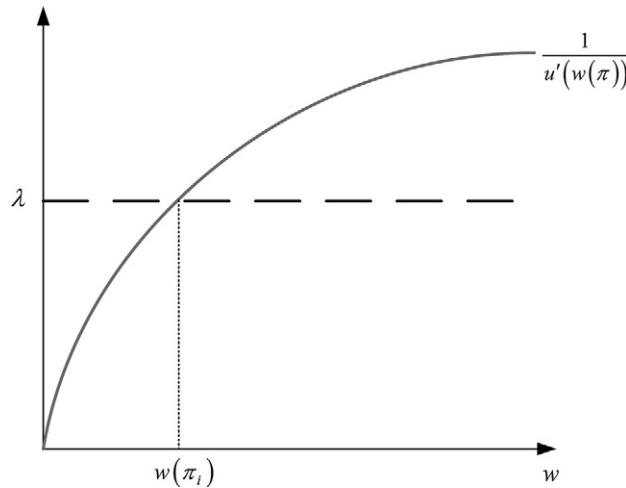


Figure 10.1
Optimal salary—Case 1

Importantly, the first-order condition in (2) entails that the principal pays a fixed wage level for all profit realizations, since

$$\lambda = \frac{1}{u'(w(\pi_i))} = \frac{1}{u'(w(\pi_j))} \quad \text{for all profits } \pi_i \neq \pi_j$$

simplifies to $u'(w(\pi_i)) = u'(w(\pi_j))$, which ultimately yields $w(\pi_i) = w(\pi_j)$ given that $u' > 0$ by definition. This is a standard risk-sharing result: the risk-neutral principal offers a contract to the risk-averse agent that guarantees the latter a fixed salary of w^* regardless of the specific profit realization that emerges.² Additionally, since the agent's PC binds, we can express it as $u(w_e^*) - g(e) = \bar{u}$, where we do not need to write it as $\sum_{i=1}^N f(\pi_i|e) u(w(\pi_i)) - g(e) = \bar{u}$ given that $w(\pi_i) = w_e^*$ for all π_i . In other words, we do not need to take the expectation of profit levels π_i as they do not affect the agent's salary (which is fixed in this case). Rearranging the PC expression above yields $u(w_e^*) = \bar{u} + g(e)$, and applying the inverse u^{-1} on both sides, we obtain a salary of

$$w_e^* = u^{-1}(\bar{u} + g(e)).$$

This expression helps us identify the salary that the principal needs to offer in order to induce a specific effort level e from the agent. In particular, for two effort levels e_L and e_H , where $e_L < e_H$, the disutility of effort function satisfies $g(e_L) < g(e_H)$, which entails

$$w_{e_L}^* = u^{-1}(\bar{u} + g(e_L)) < u^{-1}(\bar{u} + g(e_H)) = w_{e_H}^*,$$

as depicted in the figure 10.2. For instance, in order to induce e_H , we need to evaluate the utility function at a height of $\bar{u} + g(e_H)$, and the inverse $u^{-1}(\bar{u} + g(e_H))$ is graphically represented by the point in the horizontal axis that yields a height of $\bar{u} + g(e_H)$ on the vertical axis, which we find by mapping $\bar{u} + g(e_H)$ from the vertical to the horizontal axis. That is, inducing a higher effort implies offering a higher salary.

We can now plug such a salary $w_e^* = u^{-1}(\bar{u} + g(e))$ into the principal's objective function in order to find the effort level that maximizes the principal's expected profits, as follows:

$$\max_e \sum_{i=1}^N f(\pi_i|e) [\pi_i - w(\pi_i)] = \sum_{i=1}^N f(\pi_i|e) \cdot \pi_i - \underbrace{u^{-1}(\bar{u} + g(e))}_{w_e^*},$$

² However, such a salary can be a function of the effort level that the principal seeks to induce, which is an observable variable in this symmetric information setting. For this reason the fixed salary w^* is sometimes referred to as w_e^* , which emphasizes its dependence on effort levels.

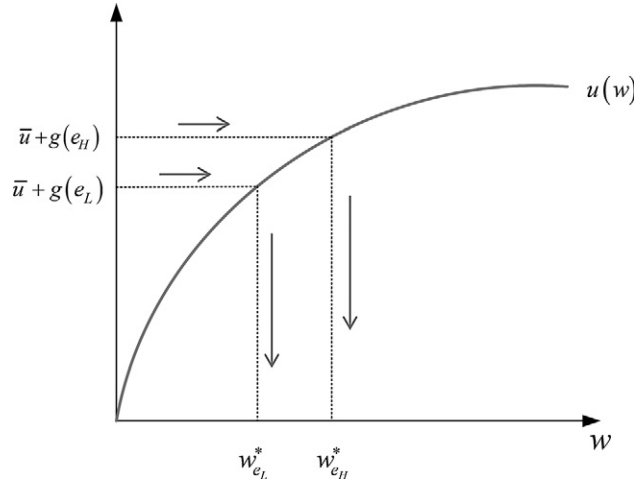


Figure 10.2
Salaries inducing two effort levels—Case 1

where the second term (salary w_e^*) does not depend on the profit realization π_i , and it is thus not multiplied by $f(\pi_i|e)$. Interestingly, this helps us reduce the number of choice variables to only the effort level e that the principal seeks to induce; that is, we no longer need the contingent list of salaries $\{w(\pi_i)\}_{i=1}^N$, since these salaries do not show up in the principal's objective function. Taking FOCs with respect to effort level e yields

$$\sum_{i=1}^N f'(\pi_i|e) \cdot \pi_i - \frac{\partial u^{-1}(\bar{u} + g(e))}{\partial e} g'(e) = 0.$$

The second term, $\partial u^{-1}(\bar{u} + g(e))/\partial e$ can be expressed as $(u^{-1})'(\bar{u} + g(e))$. By the implicit function theorem, we can rearrange it as $(u^{-1})'(\bar{u} + g(e)) = (u')^{-1}(\bar{u} + g(e))$, or $1/u'(\bar{u} + g(e))$. This notation helps us rewrite the FOC above as

$$\sum_{i=1}^N f'(\pi_i|e) \cdot \pi_i = \frac{g'(e)}{u'(\bar{u} + g(e))}. \quad (10.3)$$

Intuitively, effort is increased until the point at which its expected profit (left-hand side) coincides with its certain costs (in the right-hand side), which stems from a larger disutility of effort for the agent (numerator) that needs to be compensated with a more generous salary (denominator).

Remark We can briefly check whether the FOC in expression (3) identifies a maximum. In particular, the second-order conditions guaranteeing concavity are

$$\sum_{i=1}^N f''(\pi_i|e) \cdot \pi_i - \frac{g''(e)u'(\bar{u} + g(e)) - g'(e) \left[u''(\bar{u} + g(e)) \cdot (u^{-1})'(\bar{u} + g(e)) g'(e) \right]}{[u'(\bar{u} + g(e))]^2} \leq 0.$$

Since $(u^{-1})'(\bar{u} + g(e)) = 1/u'(\bar{u} + g(e))$, we can rewrite the expression above as

$$\sum_{i=1}^N f''(\pi_i|e) \cdot \pi_i - \frac{g''(e)}{u'(\bar{u} + g(e))} + \frac{g'(e)u''(\bar{u} + g(e))g'(e)[1/u'(\bar{u} + g(e))]}{[u'(\bar{u} + g(e))]^2} \leq 0,$$

and rearrange it as

$$\sum_{i=1}^N f''(\pi_i|e) \cdot \pi_i - \frac{g''(e)}{u'(\bar{u} + g(e))} + \frac{[g'(e)]^2 u''(\bar{u} + g(e))}{[u'(\bar{u} + g(e))]^3} \leq 0.$$

Finally, since $g'' \geq 0$ and $u' > 0$, the ratio $g''(e)/u'(\bar{u} + g(e))$ is positive; and given that $u'' \leq 0$, the last ratio of the expression above is negative (i.e., the numerator is negative but the denominator is positive). As a consequence the second-order condition is satisfied as long as the first term is not positive and significant, and it is unambiguously satisfied if the first term is negative. Intuitively, the first term is negative when the probability of higher profits, while rising in effort, does not increase at an increasing rate. That is, if the conditional probability function $f(\pi_i|e)$ is concave in effort levels.

Case 2: The principal is risk averse but the agent is risk neutral In this case the principal's benefit function $B(\pi_i - w(\pi_i))$ satisfies $B' > 0$ and $B'' < 0$, while the agent's utility function is linear in w , that is, $u(w_i - g(e)) = w_i - g(e)$. In this context the FOC under symmetric information, expression (1), becomes

$$\lambda = B'(\pi_i - w(\pi_i)),$$

since $u'(w(\pi_i)) = 1$. Unlike our results in case 1, it is now the principal who obtains a fixed payoff, which is unaffected by the profit realization π_i . Indeed, for any two profits $\pi_i \neq \pi_j$,

$$\lambda = B'(\pi_i - w(\pi_i)) = B'(\pi_j - w(\pi_j)),$$

which simplifies to $\pi_i - w(\pi_i) = \pi_j - w(\pi_j) = K$ since $B' > 0$ by definition. That is, the risk-averse principal receives the same payoff regardless of the profit realization π , whereas the risk-neutral agent now bears all the risk. In order to see this, note that in this setting, once a profit level π_i is realized, the agent only needs to make a fixed payment

K to the principal, keeping the rest of profits $\pi_i - K$ for himself. This payment scheme is therefore analogous to a franchise contract where the agent receives all profits except for a fixed fee K paid to the principal.

The agent's salary is hence $w(\pi_i) = \pi_i - K$, where K is found by making the agent indifferent between accepting and rejecting the franchise contract (i.e., the agent's PC condition must bind, as in all other cases analyzed above, since otherwise the principal could raise the fee K and still guarantee acceptance). In particular, fee K solves

$$\sum_{i=1}^N f(\pi_i|e)[\pi_i - K] - g(e) = \bar{u},$$

and solving for K yields

$$K = \sum_{i=1}^N f(\pi_i|e) \cdot \pi_i - \bar{u} - g(e).$$

In this context the principal's expected profit is

$$\sum_{i=1}^N f(\pi_i|e)[\pi_i - w(\pi_i)] = \sum_{i=1}^N f(\pi_i|e)[\pi_i - (\pi_i - K)] = \sum_{i=1}^N f(\pi_i|e) K = K,$$

where the first equality uses salary $w(\pi_i) = \pi_i - K$, and the second equality uses the property that fee $K > 0$ is independent of the profit realization. The principal's problem can then be written as

$$\max_e K = \sum_{i=1}^N f(\pi_i|e) \pi_i - \bar{u} - g(e).$$

Intuitively, the principal's profit reduces to the fixed fee K he receives from the agent, and he finds the effort level of the agent that maximizes such a fee. Taking FOC with respect to e yields

$$\sum_{i=1}^N f'(\pi_i|e) \cdot \pi_i = g'(e).$$

That is, effort is increased until the point where marginal expected profit from having the agent exert more effort (left-hand side) coincides with his marginal disutility (right-hand side). The second-order condition in this setting is more straightforward than in case 1. Specifically, we find that

$$\sum_{i=1}^N f''(\pi_i|e) \cdot \pi_i - g''(e) \leq 0.$$

Since $g''(e) \geq 0$ by definition (i.e., convex cost of effort), second-order conditions hold under the same conditions as in case 1, namely, $\sum_{i=1}^N f''(\pi_i|e)\pi_i$ must be positive but small, or such a term must be negative.

Case 3: Both the principal and the agent are risk averse This case represents our original setup. We reproduce below the FOCs with respect to w that we found earlier:

$$B'(\pi_i - w(\pi_i)) \cdot (-1) + \lambda u'(w(\pi_i)) = 0 \quad (10.4)$$

and solve for λ to get

$$\lambda = \frac{B'(\pi_i - w(\pi_i))}{u'(w(\pi_i))} \quad (10.5)$$

In order to better understand how the profit-maximizing salary is affected by the profit realization π_i , we can differentiate (10.4) with respect to w in order to apply the implicit function theorem. In particular,

$$-B''(\pi_i - w(\pi_i)) + B''(\pi_i - w(\pi_i))w'(\pi_i) + \lambda u''(w(\pi_i))w'(\pi_i) = 0.$$

Plugging the value of λ from (10.5) yields

$$-B''(\pi_i - w(\pi_i)) + B''(\pi_i - w(\pi_i))w'(\pi_i) + \frac{B'(\pi_i - w(\pi_i))}{u'(w(\pi_i))}u''(w(\pi_i))w'(\pi_i) = 0.$$

Factoring out $w'(\pi_i)$, which is the term we are interested in, yields

$$B''(\pi_i - w(\pi_i)) = \left[B''(\pi_i - w(\pi_i)) + B'(\pi_i - w(\pi_i)) \cdot \frac{u''(w(\pi_i))}{u'(w(\pi_i))} \right] w'(\pi_i).$$

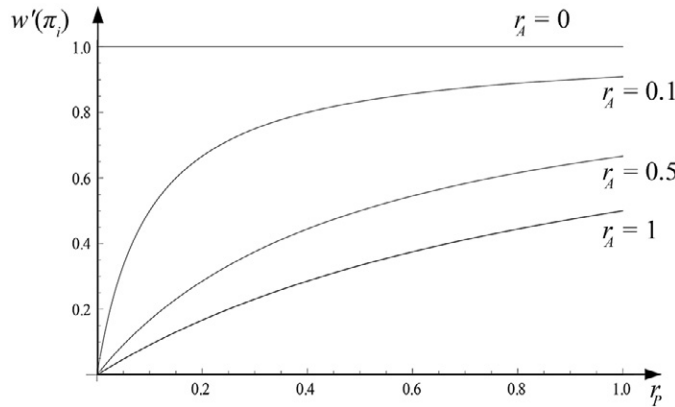
Solving for $w'(\pi_i)$, we obtain

$$w'(\pi_i) = \frac{B''(\pi_i - w(\pi_i))}{B''(\cdot) + \frac{u''}{u'} B'}.$$

We can finally divide numerator and denominator by B' , which yields

$$w'(\pi_i) = \frac{B''/B'}{B''/B' + u''/u'} = \frac{r_P}{r_P + r_A}, \quad (10.6)$$

where r_P and r_A denote the Arrow–Pratt coefficient of absolute risk aversion of the principal and the agent, respectively. As depicted in figure 10.3, the ratio in (10.6) increases in r_P , indicating that the principal provides stronger incentives to the agent as the former

**Figure 10.3**

Salary response to a higher profit realization, $w'(\pi_i)$

becomes more risk averse. Specifically, it originates at $w'(\pi_i)=0$ when $r_P=0$ (as in case 1), and increases to $w'(\pi_i)=1/(1+r_A)$ when $r_P=1$. In contrast, the ratio in (10.6) decreases in r_A , reflecting that the principal does not need to provide powerful incentives to the agent as the latter becomes more risk averse. In particular, it originates at $w'(\pi_i)=1$ when $r_A=0$ (as in case 2), and decreases reaching $w'(\pi_i)=r_P/(r_P+1)$ when $r_A=1$.

For completeness, we next evaluate the above ratio at different values of r_P and r_A :

1. *Risk neutral principal* If $r_P=0$, the principal is risk neutral, yielding $w'(\pi_i)=0$ in (10.6). (Note that this result holds regardless of the agent's coefficient or risk aversion, $r_A \geq 0$.) This setting coincides with that in case 1, where the agent receives a fixed wage; that is, he is fully insured against the profit realization π_i , whereas the risk-neutral principal bears all the risk.
2. *Risk neutral agent* If $r_A=0$, the agent is risk neutral, implying that $w'(\pi_i)=1$ in (10.4), which holds regardless of principal's coefficient of risk aversion, $r_P \geq 0$. This context coincides with that in case 2, where the risk-neutral agent bears all the risk while the principal receives a fixed payment K that insures him against different profit realization π_i , as in the franchise contract described in case 2.
3. *Agent is more risk averse than principal* If $r_A > r_P > 0$, the ratio in (10.4) is smaller than $1/2$.³ Intuitively, in this setting the agent is more risk averse than the principal, and as a consequence it is optimal for the agent's salary $w'(\pi_i)$ to exhibit small

³ To see this, note that if we divide by r_P , the ratio in (10.6) becomes $w'(\pi_i)=1/(1+(r_A/r_P))$. If $r_A > r_P > 0$, then $r_A/r_P > 1$, entailing that $1+r_A/r_P > 2$, and thus $1/(1+(r_A/r_P)) < 1/2$.

variations in the profit realization π_i . In short, the more risk-averse individual bears less payoff volatility.

4. *Principal is more risk averse than agent* If $r_A > r_P > 0$, the opposite argument applies, and the ratio in (10.6) is larger than $1/2$. In this case the less risk-averse individual (the principal) bears less risk than the agent.
5. *Same degree of risk aversion* Finally, if $r_A = r_P = r > 0$, the ratio in (10.6) collapses to $1/2$, indicating that both agents bear the same risk in the contract.

10.1.4 Asymmetric Information

When the principal cannot observe the agent's effort level, he needs to offer to the agent enough incentives to exert the profit-maximizing effort level, rather than slacking (which can be really tempting when his effort is not being observed!). How can the principal achieve this objective? He could do this by making the salary an increasing function of the realized profit. As we examine below, this is optimal even if the agent is risk averse. In particular, the principal's problem now becomes

$$\begin{aligned} & \max_{\{e, w(\pi_i)\}_{i=1}^N} \sum_{i=1}^N f(\pi_i|e) \cdot B(\pi_i - w(\pi_i)) \\ & \text{subject to } \sum_{i=1}^N f(\pi_i|e) \cdot u(w(\pi_i)) - g(e) \geq \bar{u} \quad (\text{PC}) \\ & \text{and } e^* \in \arg \max_e \sum_{i=1}^N f(\pi_i|e) \cdot u(w(\pi_i)) - g(e). \quad (\text{IC}) \end{aligned}$$

In words, the principal seeks to maximize its expected profits subject to (1) the voluntary participation of the agent (PC condition); and (2) the effort that he anticipates the agent will optimally choose in order to maximize his expected utility after receiving the contract from the principal (incentive compatibility, IC, condition).⁴

We will analyze this relatively general problem below. Now, for simplicity, let us focus on the case in which there are only two different effort levels available to the agent (e_L and e_H , where $e_H > e_L > 0$) rather than a continuum. This setting embodies the

⁴ We can intuitively understand the PC and IC conditions as if the principal puts himself in the shoes of the agent, in order to anticipate which contracts will be accepted (PC condition) and which effort level each contract induces (IC condition). Note that in this case the incentive condition IC allows the agent to choose among a continuum of effort levels, as opposed to two in previous sections. As a result the IC condition cannot be written as an expected payoff comparison, but instead as an expected utility maximization problem for the agent.

case in which the agent can choose to work a positive number of hours or completely slack from the job, namely $e_H = e > 0$ and $e_L = 0$.

In this context consider that the principal seeks to induce the high effort level e_H and that the principal is risk neutral while the agent is risk averse. Thus the principal's program reduces to

$$\begin{aligned} & \max_{\{e, w(\pi_i)\}_{i=1}^N} \sum_{i=1}^N f(\pi_i | e_H) \cdot [\pi_i - w(\pi_i)] \\ & \text{subject to } \sum_{i=1}^N f(\pi_i | e_H) u(w(\pi_i)) - g(e_H) \geq \bar{u} \end{aligned} \quad (\text{PC})$$

$$\text{and } \sum_{i=1}^N f(\pi_i | e_H) u(w(\pi_i)) - g(e_H) \geq \sum_{i=1}^N f(\pi_i | e_L) u(w(\pi_i)) - g(e_L), \quad (\text{IC})$$

where the IC condition now illustrates that the principal seeks to induce the agent to choose effort level e_H as such effort yields a higher expected utility than e_L for the agent. In this context the Lagrangean becomes

$$\begin{aligned} \mathcal{L} = & \sum_{i=1}^N f(\pi_i | e_H) \cdot [\pi_i - w(\pi_i)] + \lambda \left[\sum_{i=1}^N f(\pi_i | e_H) u(w(\pi_i)) - g(e_H) - \bar{u} \right] \\ & + \mu \left\{ \sum_{i=1}^N f(\pi_i | e_H) u(w(\pi_i)) - g(e_H) - \left[\sum_{i=1}^N f(\pi_i | e_L) u(w(\pi_i)) - g(e_L) \right] \right\}. \end{aligned}$$

Taking the first-order condition with respect to the salary w yields, for every profit realization π_i , the following equation:

$$-f(\pi_i | e_H) + \lambda f(\pi_i | e_H) u'(w(\pi_i)) + \mu [f(\pi_i | e_H) u'(w(\pi_i)) - f(\pi_i | e_L) u'(w(\pi_i))] = 0.$$

Rearranging yields

$$\underbrace{\lambda + \mu \left[1 - \frac{f(\pi_i | e_L)}{f(\pi_i | e_H)} \right]}_{\text{New}} = \frac{1}{u'(w(\pi_i))}. \quad (10.7)$$

Comparing FOC in (10.7) with that of case 1, where the principal was risk neutral but the agent risk averse in expression (10.2), we find that only the second term on the left-hand side is new. Regarding this term, first note that $\lambda > 0$ and $\mu > 0$ since both constraints bind. Intuitively, $\lambda > 0$ because, otherwise, the principal could lower the agent's salary and still guarantee his participation, and $\mu > 0$ because the principal pays

the agent the lowest salary that induces the latter to exert a high effort. Next note that $f(\pi|e_L) < f(\pi|e_H)$, which entails the probability of obtaining a given profit level π is higher when the agent exerts a high effort e_H than when he exerts a low effort e_L . As a consequence $f(\pi|e_L)/f(\pi|e_H) < 1$ and ultimately $1 - [f(\pi|e_L)/f(\pi|e_H)]$ must be a number in $(0, 1)$. Therefore the new term on the left-hand side of the FOC must satisfy

$$\lambda + \mu \left[1 - \frac{f(\pi|e_L)}{f(\pi|e_H)} \right] > \lambda.$$

Figure 10.4 depicts the FOC of this principal–agent problem under symmetric information, from expression (10.2), and under asymmetric information, from expression (10.7). Since $u'' < 0$ (and thus $1/u'$ is increasing in w), function $1/u'$ is evaluated at a higher height under asymmetric than symmetric information (see vertical axis of figure 10.4), implying that the salary the principal needs to offer when effort is not observable, w_{e_H} , is more generous than that when effort is observable, $w(\pi)$.

When deciding which effort to implement, the principal compares the effects of inducing a high effort level e_H . On one hand, effort e_H yields a positive effect on profits, since it increases the likelihood of higher profits, given that expected profits increase in effort. Importantly, this positive effect emerges under both symmetric and asymmetric information. On the other hand, inducing effort e_H also entails a negative effect, since the salary that induces such effort is higher under asymmetric than under

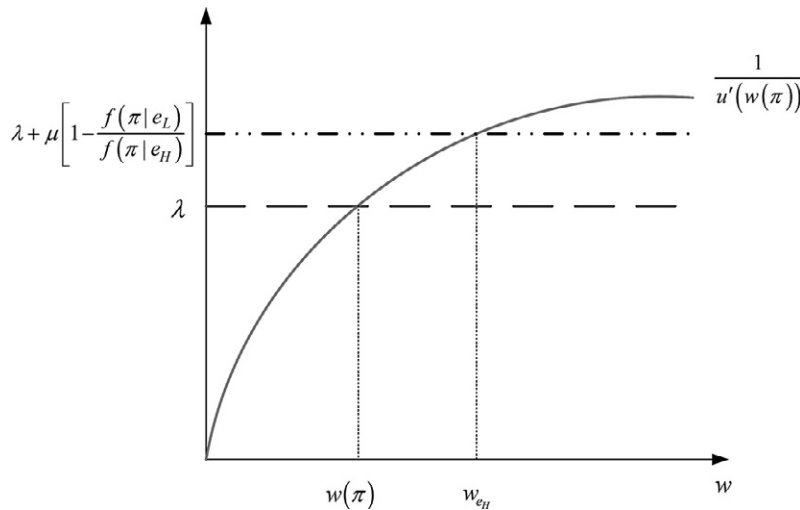


Figure 10.4
Principal–agent problem

symmetric information, that is, $w_{e_H} > w(\pi)$. In summary, since the positive effect of effort e_H arises under both information contexts, but its associated cost is larger when effort is unobservable, the principal is less willing to induce e_H when the agent's effort is unobservable than when it is observable.

10.1.5 Comparative Statics

How does the salary above change as a function of the profit realization? For that to happen, we need the left-hand side of the FOC in (10.7) to increase in π , which can only occur if the likelihood ratio $f(\pi|e_L)/f(\pi|e_H)$ decreases in π . In words, as profits increase, the likelihood of obtaining a profit level of π from effort e_H increases faster than the probability of obtaining such a profit level from e_L . (This probability is commonly known as the monotone likelihood ratio property, MLRP; described in chapter 4.) Graphically, since $\lambda + \mu[1 - (f(\pi|e_L)/f(\pi|e_H))]$ is evaluated at higher profit levels, the MLRP guarantees that $f(\pi|e_L)/f(\pi|e_H)$ declines, which produces an upward shift in the flat line of the figure 10.5, ultimately increasing the agent's wage from $w(\pi_1)$ to $w(\pi_2)$. In summary, if the likelihood ratio is monotone (i.e., it satisfies the MLRP), the principal will offer a wage compensation $w(\pi_i)$ that increases in the profit realization, π_i , since such higher observed profit makes it more likely that the agent exerted effort e_H than e_L .

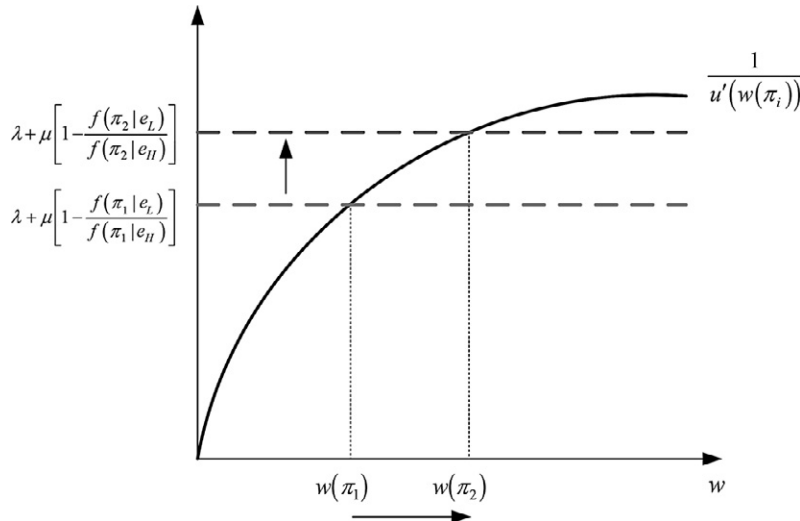


Figure 10.5
Comparative statics and MLRP

Example 10.2: Finding optimal wages under moral hazard in asymmetric information settings Let us now continue the setting presented in example 10.1, but assuming that the principal cannot observe the agent's effort. In this incomplete information setting, the principal must offer a salary that increases in profit if he seeks to induce $e_H = 5$. In particular, his maximization problem becomes

$$\begin{aligned} \max_{\{w(\pi_i)\}_{i=1}^3} \quad & 270 - [0.1w(\pi_1) + 0.3w(\pi_2) + 0.6w(\pi_3)] \\ \text{subject to} \quad & 0.1\sqrt{w(\pi_1)} + 0.3\sqrt{w(\pi_2)} + 0.6\sqrt{w(\pi_3)} - 5 \geq 9 \end{aligned} \quad (\text{PC})$$

and

$$0.1\sqrt{w(\pi_1)} + 0.3\sqrt{w(\pi_2)} + 0.6\sqrt{w(\pi_3)} - 5 \geq 0.6\sqrt{w(\pi_1)} + 0.3\sqrt{w(\pi_2)} + 0.1\sqrt{w(\pi_3)}. \quad (\text{IC})$$

Since the principal's revenue is a constant (\$270), he can alternatively minimize its expected costs (subject to the same PC and IC constraints), which helps us rewrite the program as follows:

$$\begin{aligned} \min_{\{w(\pi_i)\}_{i=1}^3} \quad & 0.1w(\pi_1) + 0.3w(\pi_2) + 0.6w(\pi_3) \\ \text{subject to} \quad & 0.1\sqrt{w(\pi_1)} + 0.3\sqrt{w(\pi_2)} + 0.6\sqrt{w(\pi_3)} - 5 \geq 9 \quad (\text{PC}) \\ & \text{and } -0.5\sqrt{w(\pi_1)} + 0.5\sqrt{w(\pi_3)} - 5 \geq 0, \quad (\text{IC}) \end{aligned}$$

where the IC constraint has been simplified. The associated Lagrangian is

$$\begin{aligned} \mathcal{L} = & 0.1w(\pi_1) + 0.3w(\pi_2) + 0.6w(\pi_3) \\ & - \lambda [0.1\sqrt{w(\pi_1)} + 0.3\sqrt{w(\pi_2)} + 0.6\sqrt{w(\pi_3)} - 5 - 9] \\ & - \mu [-0.5\sqrt{w(\pi_1)} + 0.5\sqrt{w(\pi_3)} - 5]. \end{aligned}$$

Taking FOCs with respect to wages $w(\pi_1)$, $w(\pi_2)$, and $w(\pi_3)$ yields

$$\frac{\partial \mathcal{L}}{\partial w(\pi_1)} = 0.1 - \frac{0.1\lambda}{2\sqrt{w(\pi_1)}} + \frac{0.5\mu}{2\sqrt{w(\pi_1)}} = 0, \quad (10.8)$$

$$\frac{\partial \mathcal{L}}{\partial w(\pi_2)} = 0.3 - \frac{0.3\lambda}{2\sqrt{w(\pi_2)}} = 0, \quad (10.9)$$

$$\frac{\partial \mathcal{L}}{\partial w(\pi_3)} = 0.6 - \frac{0.6\lambda}{2\sqrt{w(\pi_3)}} - \frac{0.5\mu}{2\sqrt{w(\pi_3)}} = 0. \quad (10.10)$$

Upon rearranging, (10.8) and (10.9) yield

$$\begin{aligned}\lambda &= 2\sqrt{w(\pi_2)}, \\ \mu &= 0.4\sqrt{w(\pi_2)} - 0.4\sqrt{w(\pi_1)}.\end{aligned}$$

After plugging these values into (10.10) and rearranging, we obtain

$$0.1\sqrt{w(\pi_1)} - 0.7\sqrt{w(\pi_2)} + 0.6\sqrt{w(\pi_3)} = 0. \quad (10.11)$$

Combining equation (10.11) with the (PC) and (IC) equations, we have three equations and three unknowns, namely $w(\pi_1)$, $w(\pi_2)$, and $w(\pi_3)$. The (IC) equation yields

$$w(\pi_3) = 10 + w(\pi_1).$$

Substituting this into the (PC) equation yields

$$w(\pi_2) = 36.667 - 1.33w(\pi_1).$$

Last, substituting the values of $w(\pi_2)$ and $w(\pi_3)$ in equation (10.11) gives us one equation with one unknown. With this equation we find the optimal salaries when effort is not observable:

$$w(\pi_1) = \$29.47, \quad w(\pi_2) = \$196, \quad \text{and} \quad w(\pi_3) = \$238.04.$$

Consequently the principal's expected profit would be

$$270 - [0.1 \cdot 29.46 + 0.3 \cdot 196 + 0.6 \cdot 238] = \$65.45,$$

which is lower than its profit when effort is observable (\$74). ■

10.2 Moral Hazard with a Continuum of Effort Levels—The First-Order Approach

Let us now consider a continuum of effort levels. As in previous sections (where we assumed that a worker could only have a discrete number of effort levels), the principal seeks to maximize its expected profits by anticipating the effort level that the agent selects in the second stage of the game. That is,

$$\begin{aligned} \max_{\{e, w(\pi_i)\}_{i=1}^N} \quad & \sum_{i=1}^N f(\pi_i|e) [\pi_i - w(\pi_i)] \\ \text{subject to} \quad & \sum_{i=1}^N f(\pi_i|e) u(w(\pi_i)) - g(e) \geq \bar{u} \end{aligned} \quad (\text{PC})$$

$$\text{and } e^* \in \underset{e}{\operatorname{argmax}} \sum_{i=1}^N f(\pi_i|e) u(w(\pi_i)) - g(e). \quad (\text{IC})$$

Before solving this program, let us briefly analyze how it differs relative to the problem when the agent chooses among discrete effort levels. First, note that the objective function of the principal and the PC condition for the agent coincide with those in previous sections. The agent's IC condition, however, differs as it now allows him to choose among a continuum of effort levels. Intuitively, the IC condition of the agent represents his own UMP where, for a given salary $w(\pi_i)$, the agent selects an effort level e that maximizes his expected utility. In particular, the agent differentiates his expected utility with respect to e in order to obtain

$$\sum_{i=1}^N f'(\pi_i|e) u(w(\pi_i)) - g'(e) = 0.$$

This approach, often known as the “first-order approach,” takes the agent's FOC that we just found and uses it as the IC condition in the principal's problem. This approach is valid when the agent's FOC is a sufficient condition for a maximum. That is, it is valid when agent's expected utility is concave in effort.⁵

When it is valid, we can express the principal's problem as follows:

$$\begin{aligned} & \max_{\{e, w(\pi_i)\}_{i=1}^N} \sum_{i=1}^N f(\pi_i|e) [\pi_i - w(\pi_i)] \\ \text{subject to } & \sum_{i=1}^N f(\pi_i|e) u(w(\pi_i)) - g(e) \geq \bar{u} \quad (\text{PC}) \\ \text{and } & \sum_{i=1}^N f'(\pi_i|e) u(w(\pi_i)) - g'(e) = 0, \quad (\text{IC}) \end{aligned}$$

where the agent's IC has been replaced by the FOC above (which is why we say the principal's problem uses a “first-order approach”). Then the Lagrangian becomes

⁵ In particular, the agent's second-order condition is $\sum_{i=1}^N f''(\pi_i|e) u(w(\pi_i)) - g''(e)$, which is weakly negative (thus guaranteeing concavity) if $\sum_{i=1}^N f''(\pi_i|e) u(w(\pi_i)) \leq g''(e)$. Since $g''(e) \geq 0$ by definition and $u(w(\pi_i)) \geq 0$, a sufficient condition for the agent's expected utility to be concave, and thus the first-order approach to be valid, is that the conditional probability is concave in effort, $f''(\pi_i|e) \leq 0$.

$$\begin{aligned}\mathcal{L} = & \sum_{i=1}^N f(\pi_i|e)[\pi_i - w(\pi_i)] \\ & + \lambda \left[\sum_{i=1}^N f(\pi_i|e) u(w(\pi_i)) - g(e) - \bar{u} \right] \\ & + \mu \left[\sum_{i=1}^N f'(\pi_i|e) u(w(\pi_i)) - g'(e) \right].\end{aligned}$$

Taking FOC with respect to w yields

$$\frac{\partial \mathcal{L}}{\partial w} = -f(\pi_i|e) + \lambda f(\pi_i|e) u'(w(\pi_i)) + \mu f'(\pi_i|e) u'(w(\pi_i)) = 0.$$

Dividing both sides by $f(\pi_i|e)$, we obtain

$$-1 + \lambda u'(w(\pi_i)) + \mu \frac{f'(\pi_i|e)}{f(\pi_i|e)} u'(w(\pi_i)) = 0.$$

Factoring out $u'(w(\pi_i))$ on the left-hand side and rearranging yields

$$\lambda + \mu \frac{f'(\pi_i|e)}{f(\pi_i|e)} = \frac{1}{u'(w(\pi_i))}.$$

This result is similar to that in previous sections. Since $\lambda, \mu > 0$ (recall that both PC and IC bind), the left-hand side satisfies $\lambda + \mu(f'(\pi_i|e)/f(\pi_i|e)) > \lambda$. Furthermore u' is decreasing in w (by concavity), implying that its inverse, $1/u'$, is increasing in w . Hence the principal offers a larger salary under asymmetric information than symmetric information, as depicted in figure 10.6. Finally, note that ratio $f'(\pi_i|e)/f(\pi_i|e)$ measures, in the case of continuous effort, **the likelihood ratio, which is how a marginally higher effort entails a larger probability of obtaining a given profit level π_i relative to an initial effort level.**

Let us now take FOCs with respect to e :

$$\frac{\partial \mathcal{L}}{\partial e} = \sum_{i=1}^N f'(\pi_i|e)[\pi_i - w(\pi_i)] + \mu \left[\sum_{i=1}^N f''(\pi_i|e) u(w(\pi_i)) - g''(e) \right] = 0.$$

We can rearrange this result as

$$\sum_{i=1}^N f'(\pi_i|e) \pi_i = \sum_{i=1}^N f'(\pi_i|e) w(\pi_i) - \mu \left[\sum_{i=1}^N f''(\pi_i|e) u(w(\pi_i)) - g''(e) \right]. \quad (10.12)$$

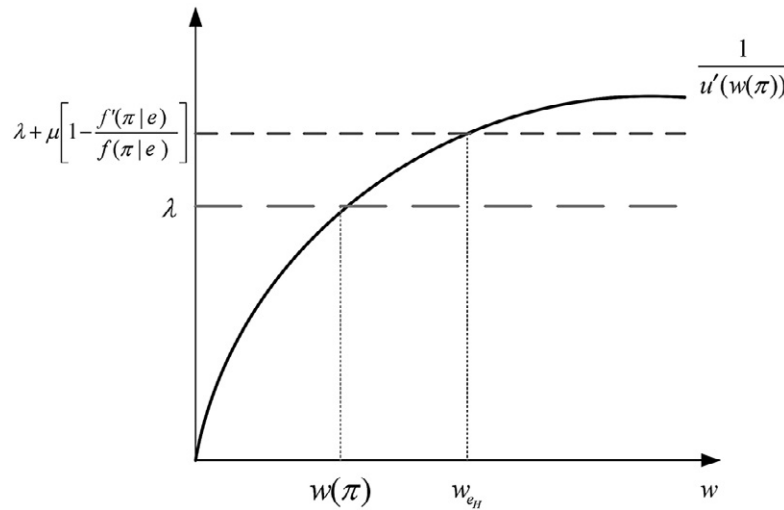


Figure 10.6
Wages in the first-order approach

In words, effort is increased until the point where its expected profits (captured on the left-hand side of equation 10.12) coincide with its associated costs (right-hand side). In particular, the cost of inducing a higher effort originates from two sources. First, a higher effort increases the probability of obtaining a higher profit, and thus the salary that the principal pays the agent once the profit is realized (see first term on the right-hand side of equation 10.12). Second, the principal must provide more incentives (higher salary) in order for the agent to exert the effort level that the principal intended (second term on the right-hand side). This last term actually reflects the fact that the agent's IC condition becomes more difficult to sustain when e increases, unless the principal offers a more generous salary.

Example 10.3: Moral hazard with continuous effort but only two possible outcomes Consider a setting in which the conditional probability satisfies the following equation: $f(\pi_i|e) = e \cdot f_H(\pi_i) + (1-e)f_L(\pi_i)$, where $e \in [0, 1]$. Intuitively, when effort is relatively high (i.e., $e \rightarrow 1$), the probability of obtaining a profit level π_i becomes $f_H(\pi_i)$, where $f_H(\pi_i) > f_L(\pi_i)$ for all π_i . In contrast, when e is close to zero, the probability of obtaining a profit of π_i declines to $f_L(\pi_i)$. In this context, the agent's expected utility is

$$EU(e) = \sum_{i=1}^N [e \cdot f_H(\pi_i) + (1-e)f_L(\pi_i)]u(w(\pi_i)) - g(e),$$

and since $e \cdot f_H(\pi_i) + (1-e)f_L(\pi_i) = e[f_H(\pi_i) - f_L(\pi_i)] + f_L(\pi_i)$, we can rewrite this expression as

$$EU(e) = \sum_{i=1}^N e \cdot [f_H(\pi_i) - f_L(\pi_i)] \mu(w(\pi_i)) + \sum_{i=1}^N f_L(\pi_i) \mu(w(\pi_i)) - g(e).$$

Before analyzing the principal's problem, we can check if the agent's expected utility is concave in effort, so we can use the first-order approach. Differencing $EU(e)$ twice with respect to effort e , we obtain $-g''(e)$, which is negative by definition (convexity of cost of effort). (Note that the first two terms of $EU(e)$ are, in this case, a linear function of e , or independent of e , respectively.) We now apply the first-order approach; in particular, the agent's FOC with respect to e is:

$$\sum_{i=1}^N [f_H(\pi_i) - f_L(\pi_i)] \mu(w(\pi_i)) = g'(e).$$

Intuitively, the agent increases his effort until the point where the increase in expected profit (and thus salary) coincides with his increase in disutility from exerting a larger effort. Plugging this FOC into the principal's problem (as the agent's IC condition), we obtain

$$\begin{aligned} & \max_{\{e, w(\pi_i)\}_{i=1}^N} \sum_{i=1}^N [e \cdot f_H(\pi_i) + (1-e)f_L(\pi_i)] \cdot [\pi_i - w(\pi_i)] \\ & \text{subject to} \quad \sum_{i=1}^N [e \cdot f_H(\pi_i) + (1-e)f_L(\pi_i)] \mu(w(\pi_i)) - g(e) \geq \bar{u} \quad (\text{PC}) \\ & \text{and} \quad \sum_{i=1}^N [f_H(\pi_i) - f_L(\pi_i)] \mu(w(\pi_i)) = g'(e). \quad (\text{IC}) \end{aligned}$$

The Lagrangian of this program is

$$\begin{aligned} \mathcal{L} = & \sum_{i=1}^N [e \cdot f_H(\pi_i) + (1-e)f_L(\pi_i)] \cdot [\pi_i - w(\pi_i)] \\ & + \lambda \left[\sum_{i=1}^N [e \cdot f_H(\pi_i) + (1-e)f_L(\pi_i)] \mu(w(\pi_i)) - g(e) - \bar{u} \right] \\ & + \mu \left[\sum_{i=1}^N [f_H(\pi_i) - f_L(\pi_i)] \mu(w(\pi_i)) - g'(e) \right]. \end{aligned}$$

Taking FOC with respect to w , we obtain the condition we found in the (more general) model above applied to this context. That is,

$$\lambda + \mu \frac{f_H(\pi_i) - f_L(\pi_i)}{e \cdot f_H(\pi_i) + (1-e)f_L(\pi_i)} = \frac{1}{u'(w(\pi_i))},$$

since $f''(\pi_i|e) = f_H(\pi_i) - f_L(\pi_i)$. Similarly, taking FOC with respect to e , we find the application of our above results to the current setting

$$\sum_{i=1}^N [f_H(\pi_i) - f_L(\pi_i)] \pi_i = \sum_{i=1}^N [f_H(\pi_i) - f_L(\pi_i)] w(\pi_i) + \mu g''(e). \blacksquare$$

Example 10.4: Moral hazard using the first-order approach Let us consider again a setting in which a principal hires an agent. In particular, the expected utility function of the agent is

$$u(w, e) = E[w] - \frac{1}{2} \rho \text{Var}[w] - c(e),$$

where ρ measures the Arrow–Pratt coefficient of absolute risk aversion, and $e \in [0, 1]$ measures his effort, where the cost of effort is given by $c(e) = \frac{1}{2}e^2$. (Recall that CARA utility functions, such as $u(w) = -\exp^{-\rho w}$, exhibit a constant Arrow–Pratt coefficient of risk aversion, given that $r_A(w) \equiv -u''(w)/u'(w) = \rho$.) The outcome of the project, x , is stochastic and given by

$$x = f(e, \varepsilon) = e + \varepsilon, \quad \text{where } \varepsilon \sim N(0, \sigma^2).$$

The agent's reservation utility is $\bar{u} = 0$. The principal offers a linear contract to the agent

$$w(x) = a + bx,$$

where parameter $a > 0$ can be interpreted as a fixed payment, and $b \in [0, 1]$ denotes the share of profits that the agent receives (bonus). In this context, expected profits are

$$\begin{aligned} E[\pi] &= E[x - w] = E[x] - E[w] \\ &= E[x] - (a + bE[x]) = (1 - b)e - a. \end{aligned}$$

Since $E[x] = e$, the expected utility of the agent when he exerts effort level e is

$$\begin{aligned} E[u(w, e)] &= E[w] - \frac{1}{2}\rho \text{Var}[w] - c(e) \\ &= a + be - \frac{1}{2}\rho b^2 \sigma^2 - \frac{1}{2}e^2, \end{aligned}$$

given that $E[w] = a + be$, $\text{Var}[x] = b^2 + \sigma^2$, and $c(e) = \frac{1}{2}e^2$. Taking FOC with respect to e , we can find the effort that the agent chooses⁶

$$\frac{\partial E[u(w, e)]}{\partial e} = b - e = 0,$$

implying that the agent chooses an effort level of $e = b$. Hence the principal's problem is to choose the fixed payment, a , and the bonus, b , to solve

$$\max_{\{e, a, b\}} (1 - b)e - a$$

$$\text{subject to } a + be - \frac{1}{2}\rho b^2 \sigma^2 - \frac{1}{2}e^2 \geq 0 \quad (\text{PC})$$

$$\text{and } e = b. \quad (\text{IC})$$

First, we plug the result that we found from agent's IC, $e = b$, into the program above. Second, we simplify the PC to a more compact expression, $a + \frac{1}{2}b^2(1 - \rho\sigma^2) \geq 0$, which yields a problem with only two choice variables and only one constraint:

$$\max_{\{a, b\}} (1 - b)b - a$$

$$\text{subject to } a + \frac{1}{2}b^2(1 - \rho\sigma^2) \geq 0, \quad (\text{PC})$$

with the associated Lagrangian,

$$\mathcal{L} = (1 - b)b - a + \lambda \left[a + \frac{1}{2}b^2(1 - \rho\sigma^2) \right].$$

⁶ We use the first-order approach since differentiating $b - e$ again with respect to e yields $-1 \leq 0$, implying that the second-order condition holds. For examples of principal-agent problems in which the first-order approach is not valid (because FOCs produce a minimum, rather than a maximum); see Bolton and Dewatripont (2005, ch. 4).

The Kuhn–Tucker conditions are

$$\frac{\partial \mathcal{L}}{\partial a} = -1 + \lambda = 0 \rightarrow \lambda = 1, \quad (10.13)$$

$$\frac{\partial \mathcal{L}}{\partial b} = 1 - 2b + \lambda b(1 - \rho\sigma^2) = 0, \quad (10.14)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = a + \frac{1}{2}b^2(1 - \rho\sigma^2) = 0. \quad (10.15)$$

From (10.13), $\lambda = 1$. Plugging this result into (10.14) yields

$$1 - 2b + b(1 - \rho\sigma^2) = 0,$$

and solving for the bonus b , we obtain $b = 1/(1 + \rho\sigma^2)$, which is decreasing in the agent's risk aversion parameter, ρ , and in the variance of outcomes, σ^2 .

Let us now find the remaining unknown, namely the fixed payment a . We can find it by using agent's participation constraint, which binds since $\lambda = 1$. In particular, plugging $b = 1/(1 + \rho\sigma^2)$ into the (binding) PC yields

$$a + \frac{1}{2} \left(\frac{1}{1 + \rho\sigma^2} \right)^2 (1 - \rho\sigma^2) = 0.$$

And solving for the fixed payment a yields

$$a = -\frac{1}{2} \left(\frac{1}{1 + \rho\sigma^2} \right)^2 (1 - \rho\sigma^2).$$

For illustration purposes, let us analyze the extreme cases first. If $\sigma^2 = 0$, effort is deterministic (a perfect predictor of profits), whereby $x = f(e) = e$, so then a and b become

$$b = \frac{1}{1 + \rho \cdot 0} = 1 \quad \text{and} \quad a = -\frac{1}{2} \left(\frac{1}{1 + \rho \cdot 0} \right)^2 (1 - \rho \cdot 0) = -\frac{1}{2}.$$

Intuitively, the agent gives the principal a fixed payment, keeping the remaining profits, thus benefiting from high-powered incentives, as in franchises. If, in contrast, $\sigma^2 = 1$, effort is a very imprecise predictor of outcomes, so then

$$b = \frac{1}{1 + \rho} \quad \text{and} \quad a = \frac{1}{2} \left(\frac{1}{1 + \rho} \right)^2 (1 - \rho).$$

(Note that the fixed payments, a , and the bonus, b , are decreasing in the agent's risk aversion parameter, ρ .) More generally, there is a negative relationship between a and b as σ^2 increases: (1) when σ^2 is low (i.e., all effort levels yield a similar outcome x), the fixed payment a is low (or even negative) while the bonus b is high; and (2) when σ^2 is high, the fixed payment a is high (though it can still be negative) whereas the bonus b is low. ■

10.3 Moral Hazard with Multiple Signals

Consider a setting in which the principal, still not observing effort e , observes not only the profits π of the firm but also a signal s , based on a middle management report about the agent's performance. Such a signal provides no intrinsic economic value (i.e., s does not affect profits), but it does provide information about effort e . In earlier problems, the probability density function had only one argument, π . In this problem, the probability density function has two observables, π and s . Setting up the problem the same way as in the asymmetric information section, yielding equation (10.7), we obtain the condition

$$\frac{1}{u'(w)} = \gamma + \mu \left[\frac{f(\pi, s|e_L)}{f(\pi, s|e_H)} \right].$$

Hence variations in s affect wages only if $f(\pi, s|e) \neq f(\pi|e)$, that is, if π is not a sufficient statistic of e . Intuitively, the pair (π, s) contains more information about the manager's exerted effort e than π alone. If $f(\pi, s|e) = f(\pi|e)$, then the observation of signal s becomes uninformative (provides no more information than π alone), and the problem of asymmetric information, and its results in equation (10.7) would coincide with those in the current case.

As in the comparative statics section above, where we analyzed whether salary w was increasing in the observed profits π , we can now apply a similar approach to examine under which conditions w increases in signal s . Specifically, for two signals s_1 and s_2 where $s_2 > s_1$, if salary increases in the signal, $w(\pi, s_2) > w(\pi, s_1)$, we would have that $u'(w)$ decreases (by concavity), which in turn increases $1/u'(w)$, on the left-hand side of the FOC above. Hence the right-hand side must also increase in s , that is,

$$\gamma + \mu \left[1 - \frac{f(\pi, s_2|e_L)}{f(\pi, s_2|e_H)} \right] > \gamma + \mu \left[1 - \frac{f(\pi, s_1|e_L)}{f(\pi, s_1|e_H)} \right].$$

As depicted in figure 10.7, this inequality ultimately requires that

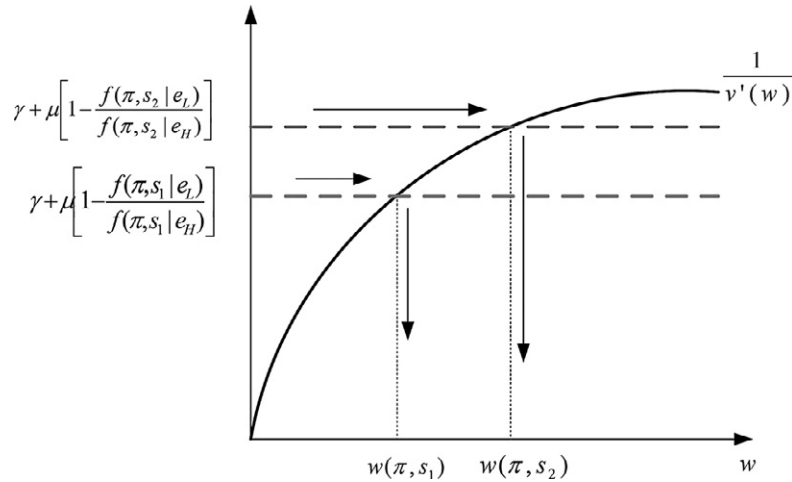


Figure 10.7
Moral hazard with multiple signals

$$\frac{f(\pi, s_2 | e_L)}{f(\pi, s_2 | e_H)} < \frac{f(\pi, s_1 | e_L)}{f(\pi, s_1 | e_H)},$$

or, after cross-multiplying,

$$\frac{f(\pi, s_1 | e_H)}{f(\pi, s_1 | e_L)} < \frac{f(\pi, s_2 | e_H)}{f(\pi, s_2 | e_L)}.$$

In words, for the salary to increase in the intermediate signal s that the principal receives, we need such a signal to have a decreasing likelihood ratio. That is, a high effort is relatively more likely to originate from a high than from a low signal (i.e., more likely to come from s_2 than from s_1).

Extensions The moral hazard model we analyzed in this section can be extended in many dimensions: (1) allowing for the firm manager to have a team of employees (not observing the effort exerted by any of them), as analyzed by Holstrom (1982), Mookherjee (1984), and Itoh (1991), and examined in exercise 9; (2) permitting the agent to exert effort on different, rather than a single, tasks (e.g., producing goods and commercializing them), as described by Holstrom and Milgrom (1991); and (3) allowing dynamic contracts lasting for several periods, as in Bolton and Dewatripont (2005, ch. 10) and references therein. Finally, for empirical work testing the presence of moral hazard in different markets, see Salanié (1997, ch. 8).

10.4 Adverse Selection—The “Lemons” Problem

As described in previous chapters, under symmetric information markets often work well, and allow buyers interested in a particular good to find sellers offering that good, ultimately purchasing it at a mutually agreeable price. Under asymmetric information, however, markets do not necessarily work well (as we next show, they might even fail to exist!) even when both buyers and sellers would benefit from their existence.

Following the original model of Akerloff (1970), consider a market of used cars, whose quality is denoted by q , where $q \sim U[0, Q]$ and $Q \in (1, 2)$. A car of quality q is valued as such by the buyer, and as q/Q by the seller. Since $q/Q < q$, the buyer assigns a higher value to the car than the seller, thus allowing both parties to exchange the car at a price p between q/Q and q and make a profit (for the seller) and a surplus (for the buyer). In particular, if car of quality q is exchanged at price p the buyer obtains a utility

$$u(p, q) = q - p,$$

while the seller makes a profit of

$$\pi(p, q, Q) = p - \frac{q}{Q}.$$

Assume that there are a sufficient number of buyers prompting all gains from trade to be appropriated by the seller.

10.4.1 Symmetric Information

When the buyer can perfectly observe the car quality q , he buys at a price p if and only if his utility from such a trade is positive, $q - p \geq 0$, or $p \leq q$. Anticipating such a response from the buyer, a seller with a car of quality q sets a price that solves

$$\max_{p \geq 0} p - \frac{q}{Q}$$

$$\text{subject to } p \leq q, \tag{PC}$$

where $p \leq q$ can be interpreted as the buyer’s participation constraint (PC), meaning that for prices $p > q$ he would reject the price, and the trade would not take place. Condition (PC) must bind; otherwise, the seller could increase the price p and extract a larger profit from the buyer. We thus have $p = q$, which can be plugged into the seller’s objective function (second term) to yield the following unconstrained problem:

$$\max_{p \geq 0} p - \frac{p}{Q}.$$

Taking the FOC with respect to p yields $1 - (1/Q)$, or $(Q-1)/Q > 0$. Since $Q > 1$ by definition, a corner solution exists whereby the seller raises the price p as much as possible (as long as it is accepted by the buyer). Hence the price under symmetric information is $p^{SI} = q$ for every quality q .

10.4.2 Asymmetric Information

When the buyer is unable to observe the car's true quality q , he forms an expectation $E[q]$. Hence the buyer accepts a trade if the car's asking price p satisfies $p \leq E[q]$. In this context the seller anticipates such an acceptance rule by the buyer and sets a price p that solves

$$\max_{p \geq 0} p - \frac{p}{Q}$$

$$\text{subject to } p \leq E[q]. \quad (\text{PC})$$

By the same argument as under symmetric information, PC binds entailing $p = E[q]$, which becomes the price that the seller sets as long as his profits are positive, that is,

$$p - \frac{q}{Q} = E[q] - \frac{q}{Q} \geq 0 \Leftrightarrow q \leq Q \cdot E[q],$$

as depicted in figure 10.8, where cars with quality to the left of $Q \cdot E[q]$ are offered by the seller, while those exceeding this quality level are not profitably offered (and its market ceases to exist).

For instance, when quality q is uniformly distributed, that is, $q \sim U[0, Q]$, its expected value becomes

$$E[q] = \frac{Q-0}{2} = \frac{Q}{2},$$

and $Q \cdot E[q]$ is then $Q^2/2$.⁷ Hence all cars with relatively low quality $q \leq Q^2/2$ are offered by the seller at a price $p = E[q] = Q/2$, yielding a profit of $Q/2 - q/2$ for the seller,

⁷ For instance, if $Q = 3/2$ since $Q \in (1, 2)$, then the buyer's expected value is $(3/2)^2/2 = 3/4$, and cutoff $Q \cdot E[q]$ of cars offered by the seller becomes $(3/2) \cdot 3/4 = 9/8$, leaving cars in the interval $(9/8, 3/2)$ not offered by the seller.

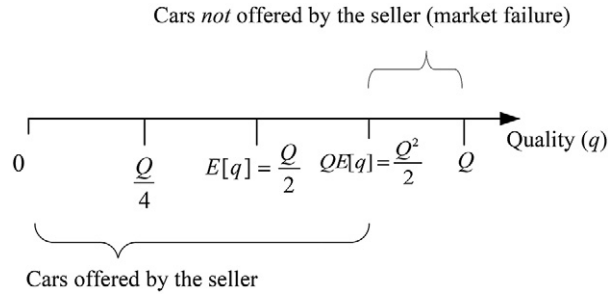


Figure 10.8
Lemons problem

and a zero (expected) utility for the buyer since $p = E[q]$. However, cars with relatively high quality, $q > Q^2/2$, are not offered by the seller since the highest price he can charge to the uninformed buyer, $p = E[q]$, does not compensate the seller's costs. This is problematic because, as stated above in the symmetric information model, seller and buyer could make a positive margin from exchanging cars of all quality levels if they could both observe their quality. In other words, the buyer's inability to observe q leads to the non-existence of the market for good cars ("peaches"), whereas only bad cars ("lemons") exist in the market.

10.4.3 Further Iterations

The previous result assumes that the buyer forms his expectation of the average car quality, $E[q]$, and the seller, anticipating such a lack of information, sets a price $p = E[q]$. However, one could argue that a fully rational buyer would anticipate such a pricing decision by the seller and, hence, that the seller finds it worthy to only offer low-quality cars, namely those with $p \leq Q^2/2$. In that case, the buyer anticipates that only cars of quality $q \in [0, Q^2/2]$ are offered, and if quality is uniformly distributed $q \sim U[0, Q]$, buyers can compute the expected quality of those offered cars as

$$E\left[q \mid q \leq \frac{Q^2}{2}\right] = \frac{(Q^2/2) - 0}{2} = \frac{Q^2}{4}.$$

Therefore the buyer would only buy cars whose price satisfies $p \leq Q^2/4$. The seller would then set the price at $p = Q^2/4$, yielding a profit of $p - q/Q = Q^2/4 - q/Q$ which is positive only if quality q satisfies $q \leq Q^3/4$.

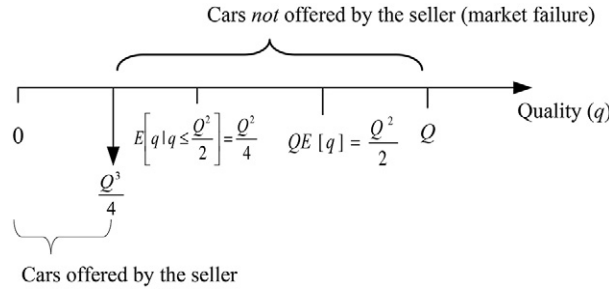


Figure 10.9
Unraveling

Figure 10.9 illustrates that the seller now further restricts the set of cars that he offers, moving toward cars of worse average quality (note that $Q^3/4 < Q^2/4 < Q^2/2$ given that $Q \in (1, 2)$ by assumption).

The argument above can, of course, be extended since now a rational buyer would update its expected car quality (among those offered by the seller) to those satisfying $q \leq Q^3/4$, yielding an expected quality of only

$$E\left[q \mid q \leq \frac{Q^3}{4}\right] = \frac{(Q^3/4) - 0}{2} = \frac{Q^3}{8},$$

leading the seller to offer cars that yield a positive profit, that is, those with quality q satisfying $Q^3/8 - q/Q \geq 0$, or $q \leq Q^4/8$, which lies closer to zero than cutoff $Q^3/4$ in figure 10.9. In summary, the seller would shift the set of offered cars even more to the left of the quality line toward worse cars (closer to zero). Repeating the same argument enough times, we find that the market “unravels,” as it only offers cars of the worse possible quality, $q=0$, and the buyer is only willing to pay a price of $p=0$, leaving all other types of cars unsold.

Example 10.5: Lemons problem Consider a market of used cars with maximum available quality $Q=1.9$, and that q is uniformly distributed in $[0, Q]$. Recall that $Q \in (1, 2)$, so our example illustrates the availability of several cars of relatively good quality. In this context, the buyer’s expected value is $1.9/2=0.95$, and the cutoff $Q \cdot E[q]$ of cars offered by the seller becomes $1.9 \cdot 0.95=1.805$, leaving cars in the interval $(1.805, 1.9)$ unoffered. Under a context of complete information, these cars would have been bought by the buyer who values them at q , and sold by the seller who values them at only $q/1.9=0.52q$, but the presence of asymmetric information between buyer and seller prevents mutually beneficial trades from occurring.

Additionally a rational buyer will anticipate that only cars in the interval $(0, 1.805)$ are offered by the seller, and thus update his expected value of offered cars to $E[q|q \leq 1.805] = (1.805 - 0)/2 = 0.9$, which leads the seller to only offer those cars with quality $q \leq 0.9^3/4 = 1.71$. The set of offered cars is thus restricted from $(0, 1.805)$ to $(0, 1.71)$. A similar argument applies to further iterations in the buyer's expected car quality. ■

Application to Labor Markets

We can extend a similar argument to other markets where buyers have access to less information than sellers. For instance, consider a competitive labor market with many firms seeking to hire a worker for a specific position. The worker (seller of labor services) privately observes his own productivity θ , but firms (the buyer of labor in this context) cannot observe it. In this setting, firms would offer a wage according to the worker's expected productivity, which is $E[\theta] = 1/2$ in the case that $\theta \sim U[0, 1]$. For this salary, only workers with a productivity lower than $1/2$ would be interested in accepting the position, while those with $\theta > 1/2$ will be left unemployed. The argument continues since a fully rational manager will only offer a salary of $w = E[\theta|\theta \leq 1/2] = 1/4$, which is the expected productivity of the subset of workers accepting the job, entailing that only those workers with productivity $\theta \leq 1/4$ accept the job. Extending the argument infinite times, we find that workers with lowest productivity level $\theta = 0$ are employed, while the labor market for all other worker types $\theta > 0$ unravels (i.e., it ceases to exist due to the firm's asymmetric information).

Solutions to the Adverse Selection Problem

The market failure described above can be overcome by a number of tools. First, sellers can offer warranties for their cars in order to signal their quality. In that setting we seek to find under which conditions a separating equilibrium exists, whereby lemons are sold without warranty while peaches are sold with warranty, and all cars are exchanged at their symmetric information price.⁸ We explore this type of separating equilibrium in the exercises at the end of the chapter. Specifically, we show that, if offering a warranty is more costly for the seller of lemons than for the one selling peaches, a separating equilibrium emerges where warranties serve as signals that sellers use to convey their private information to uninformed buyers, who can now easily infer the car's quality. A second solution to the lemons problem is screening, whereby the

⁸ Rather than offering warranties, the seller could sell the car as a certified pre-owned (CPO) vehicle, which go through a more thorough inspection and repair service at the dealership than used cars do. Customers seem to interpret these cars have a better expected quality than similar used cars, since casual observation indicates CPOs are often sold at higher prices.

principal (in this case the buyer) offers a menu of contracts to the agent (the seller), to induce each type of agent to voluntarily choose the contract meant for his type, whereby the contracts induce self-selection. Screening is often easier to interpret in the standard principal–agent relationship between a firm manager and a worker. In the next section we present the principal–agent model, and then examine how to design screening contracts in that setting.

10.5 Adverse Selection—The Principal–Agent Problem

Consider a setting where a firm (the principal) seeks to hire a worker (an agent) but cannot observe the worker's cost of effort, which ultimately affects the amount of effort that the worker exerts and thus the firm's profits. Hence the firm's manager would like to know the worker's cost of effort in order to design his salary. In particular, the firm's profit function is

$$\pi(e, w) = x(e) - w,$$

where function $x(e)$ represents the benefit that the manager obtains when the worker supplies e units of effort, which is increasing and concave in effort, $x'(e) \geq 0$ and $x''(e) \leq 0$. Now the worker's utility function is

$$v(w, e | \theta) = u(w) - c(e, \theta),$$

where $u(w)$ denotes the value he assigns to salary w , $u'(w) > 0$, $u''(w) \leq 0$; and $c(e, \theta)$ represents the worker's cost of exerting e units (e.g., hours) of effort when his type is θ . For simplicity, we assume that the worker can only be of two types, θ_L and θ_H , where $\theta_L < \theta_H$. Intuitively, a high-type worker faces a higher total and marginal cost of effort, expressed as $c(e, \theta_L) < c(e, \theta_H)$ and $c'(e, \theta_L) < c'(e, \theta_H)$, respectively, for every $e > 0$.

For comparison purposes, we first analyze a symmetric information setting in which the principal can observe the agent's type, and evaluate the optimal contract of salary and exerted effort (w_i, e_i) , where $i = (H, L)$ denotes the worker's type. In the following subsection we examine a more realistic setting where the principal cannot observe the agent's type but knows that the relative frequency of low-type workers in the population is $p \in (0, 1)$ while that of the high-type workers is $1 - p$. In such circumstances the principal will need to design a contract different from that under symmetric information. Otherwise, workers with a low cost of effort could lie about their true type in order to benefit from the contract meant for the worker with a high cost of effort. We show that the principal designs a menu of contracts, (w_L, e_L) for the low type and (w_H, e_H) for the high type, which induces each type of worker to choose the contract meant for him (self-selection of contracts).

10.5.1 Symmetric Information

If the principal (firm) knew that the agent is type $i = \{H, L\}$, it would solve

$$\begin{aligned} \max_{w_i, e_i} \quad & x(e_i) - w_i \\ \text{subject to} \quad & u(w_i) - c(e_i, \theta_i) \geq 0, \end{aligned} \quad (\text{PC})$$

where the (PC) constraint guarantees that the worker willingly accepts the contract. Since the firm can reduce w_i until (PC) holds with equality, (PC) must bind, implying that

$$u(w_i) = c(e_i, \theta_i),$$

or, after inverting $u(\cdot)$ on both sides, $w_i = u^{-1}[c(e_i, \theta_i)]$. This result helps us simplify the problem above to an unconstrained maximization problem with only one choice variable (the effort level), as follows:

$$\max_{e_i} x(e_i) - u^{-1}[c(e_i, \theta_i)],$$

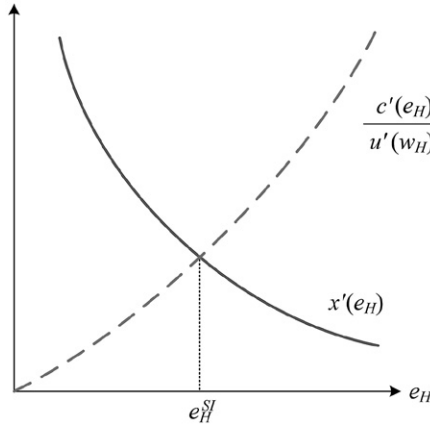
since $w_i = u^{-1}[c(e_i, \theta_i)]$. Taking FOC with respect to e_i yields

$$x'(e_i) = \frac{1}{u'(u^{-1}[c(e_i, \theta_i)])} c'(e_i, \theta_i),$$

or more compactly,

$$x'(e_i) = \frac{c'(e_i, \theta_i)}{u'(w_i)}.$$

Hence effort is increased until the point at which the marginal rate of substitution of effort and wage for the firm (left-hand side, since $MRS_{e,w} = MU_e / MU_w = x'(e)$) coincides with that of the worker (right-hand side). Intuitively, the firm's MRS in this context illustrates how much the worker's effort needs to increase after a given \$1 increase in wages so that the firm's profits are unaffected. A similar argument applies to the worker, where his MRS represents how much he can increase effort after receiving a \$1 increase in the salary so that his utility remains constant. The equality between the principal's and the agent's MRS under symmetric information reveals that increasing/decreasing effort is not Pareto improving. Figure 10.10 depicts the left-hand side of the above FOC, $x'(e_i)$, which is decreasing since $x''(e_i) \leq 0$ by definition; and the right-hand side, $c'(e_i, \theta_i) / u'(w)$, which is increasing since $c''(e_i) \geq 0$ by definition. Their crossing point identifies the profit-maximizing effort under symmetric information, e_i^{SI} .

**Figure 10.10**

Principal-agent problem under symmetric information

Example 10.6: Principal-agent problem under symmetric information Consider a principal and an agent where the probability of facing a low type is $p = \frac{1}{2}$, types are $\theta_L = 1$, $\theta_H = 2$, productivity of effort is $x(e) = \log(e)$, and $u(w) = w$. The cost of effort is $c(e, \theta) = \theta e^2$, implying that the marginal cost of effort is thus $2\theta e$, which is positive and increasing in e . Therefore the principal's profit function is $\pi(w, e) = \log(e) - w$ and the agent's utility is $v(w, e|\theta_i) = w - \theta_i e^2$.

Under symmetric information, the FOC we derived earlier becomes

$$x'(e_i) = \frac{c'(e_i, \theta_i)}{u'(w)} \Rightarrow \frac{1}{e_i} = \frac{2\theta_i e}{1},$$

and solving for e_i , we obtain

$$e_i^2 = \frac{1}{2\theta_i} \rightarrow e_i^{SI} = \left(\frac{1}{2\theta_i} \right)^{1/2}.$$

In order to find the optimal salary, we can use the (PC), which holds with equality, $u(w_i) = c(e_i, \theta_i)$. Since in this context $u(w_i) = w_i$ and $c(e_i, \theta_i) = \theta_i e_i^2$, we obtain a salary of

$$w_i = \theta_i (e_i^{SI})^2 = \theta_i \left(\left(\frac{1}{2\theta_i} \right)^{1/2} \right)^2 = \frac{1}{2}.$$

Plugging in our values for θ_L and θ_H , we find optimal contracts of

$$(w_H^{SI}, e_H^{SI}) = \left(\frac{1}{2}, \frac{1}{2}\right) = (0.5, 0.5),$$

$$(w_L^{SI}, e_L^{SI}) = \left(\frac{1}{2}, \frac{1}{\sqrt{2}}\right) \approx (0.5, 0.707).$$

Intuitively, the firm will pay both types of workers the same wage under symmetric information, but expect a higher effort level from the low-cost worker, $e_L^{SI} > e_H^{SI}$. ■

10.5.2 Asymmetric Information

When the firm cannot observe the worker's type, it seeks to maximize the expected profits by designing a pair of contracts, (w_H, e_H) and (w_L, e_L) , that satisfy four constraints: (1) voluntary participation of the high-type worker; (2) voluntary participation of the low-type worker; (3) that the high-type worker prefers the contract meant for him, (w_H, e_H) rather than that for the low-type, (w_L, e_L) ; and (4) that the low-type worker prefers contract (w_L, e_L) rather than that for the high-type worker (w_H, e_H) . Since every type of worker has an incentive to select the contract meant for him, we say that these contracts induce “self-selection.” Hence the firm solves the following profit-maximization problem

$$\begin{aligned} \max_{w_L, e_L, w_H, e_H} \quad & p[x(e_L) - w_L] + (1-p)[x(e_H) - w_H] \\ \text{subject to} \quad & u(w_H) - c(e_H, \theta_H) \geq 0 & (PC_H), \\ & u(w_L) - c(e_L, \theta_L) \geq 0 & (PC_L), \\ & u(w_H) - c(e_H, \theta_H) \geq u(w_L) - c(e_L, \theta_H) & (IC_H), \\ & u(w_L) - c(e_L, \theta_L) \geq u(w_H) - c(e_H, \theta_L) & (IC_L), \end{aligned}$$

where the first constraint, PC_H , represents the participation constraint for the high-type worker; the second constraint, PC_L , denotes that for the low-type worker; the third constraint, IC_H , is the incentive compatibility condition for the high-type worker so he does not have incentives to choose the contract meant for the low-type worker; and the fourth constraint, IC_L , is the incentive compatibility condition for the low-type worker so that he prefers the contract meant for him rather than that of the high-type worker. As described above, the PC constraints guarantee voluntary participation of all types of workers, while the IC constraints ensure self-selection.

Before finding Kuhn–Tucker conditions, note that PC_L is implied by IC_L and PC_H since

$$u(w_L) - c(e_L, \theta_L) \geq u(w_H) - c(e_H, \theta_L) > u(w_H) - c(e_H, \theta_H) \geq 0,$$

where the first (weak) inequality originates from IC_L ; the second (strict) from the assumption on the cost of effort, namely $c(e, \theta_L) < c(e, \theta_H)$ for every e ; and the third (weak) inequality stems from the PC_H condition. As a consequence we find that $u(w_L) - c(e_L, \theta_L) > 0$, as required by PC_L . Therefore PC_L holds with strict inequality whereas that of the least productive agent, PC_H , binds (which we prove below).

The Lagrangian of the preceding maximization problem is

$$\begin{aligned} \mathcal{L} = & p[x(e_L) - w_L] + (1-p)[x(e_H) - w_H] \\ & + \lambda_1[u(w_H) - c(e_H, \theta_H)] \\ & + \lambda_2[u(w_H) - c(e_H, \theta_H) - u(w_L) + c(e_L, \theta_H)] \\ & + \lambda_3[u(w_L) - c(e_L, \theta_L) - u(w_H) + c(e_H, \theta_L)], \end{aligned}$$

where PC_L is not considered since it holds with strict inequality, as previously shown. Taking FOCs, we obtain

$$\frac{\partial \mathcal{L}}{\partial w_L} = -p - \lambda_2 u'(w_L) + \lambda_3 u'(w_L) = 0,$$

$$\frac{\partial \mathcal{L}}{\partial w_H} = -(1-p) + \lambda_1 u'(w_H) + \lambda_2 u'(w_H) - \lambda_3 u'(w_H) = 0,$$

$$\frac{\partial \mathcal{L}}{\partial e_L} = px'(e_L) + \lambda_2 c'(e_L, \theta_L) - \lambda_3 c'(e_L, \theta_L) = 0,$$

$$\frac{\partial \mathcal{L}}{\partial e_H} = (1-p)x'(e_H) - \lambda_1 c'(e_H, \theta_H) - \lambda_2 c'(e_H, \theta_H) + \lambda_3 c'(e_H, \theta_L) = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_1} = u(w_H) - c(e_H, \theta_H) \geq 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_2} = u(w_H) - c(e_H, \theta_H) - u(w_L) + c(e_L, \theta_H) \geq 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_3} = u(w_L) - c(e_L, \theta_L) - u(w_H) + c(e_H, \theta_L) \geq 0,$$

and $\lambda_i \geq 0$ for all $i \in \{1, 2, 3\}$. For simplicity, we next consider that the cost of effort takes the form

$$c(e, \theta_i) = \theta_i c(e) \quad \text{for all } i = \{H, L\},$$

where $c(e)$ is increasing and convex in effort, $c'(e) \geq 0$ and $c''(e) \geq 0$. (Note that such function satisfies the initial conditions of total and marginal cost of effort being higher for the high than for the low type, i.e., $\theta_H c(e) > \theta_L c(e)$ and $\theta_H c'(e) > \theta_L c'(e)$ for all $e > 0$.) Rearranging, the first two FOCs yields

$$-\lambda_2 + \lambda_3 = \frac{p}{u'(w_L)},$$

$$\lambda_1 + \lambda_2 - \lambda_3 = \frac{1-p}{u'(w_H)},$$

then adding them together yields

$$\lambda_1 = \frac{p}{u'(w_L)} + \frac{1-p}{u'(w_H)}.$$

Therefore, $\lambda_1 > 0$, implying that the constraint associated with Lagrange multiplier λ_1 , PC_H , binds, that is, $u(w_H) - c(e_H, \theta_H) = 0$. The third FOC can be rewritten as

$$px'(e_L) = \lambda_3 \theta_L c'(e_L) - \lambda_2 \theta_H c'(e_L),$$

and, after rearranging, we get

$$\frac{px'(e_L)}{c'(e_L)} = \lambda_3 \theta_L - \lambda_2 \theta_H.$$

And the fourth FOC can be written as

$$(1-p)x'(e_H) = \lambda_1 \theta_H c'(e_H) - \lambda_3 \theta_L c'(e_H) + \lambda_2 \theta_H c'(e_H),$$

which we can rearrange as

$$\frac{(1-p)x'(e_H)}{c'(e_H)} = \lambda_1 \theta_H - (\lambda_3 \theta_L - \lambda_2 \theta_H).$$

Combining the two (rearranged) FOCs yields

$$\frac{(1-p)x'(e_H)}{c'(e_H)} = \lambda_1 \theta_H - \frac{px'(e_L)}{c'(e_L)}$$

Solving for $\lambda_1\theta_H$, and using $\lambda_1 = p/u'(w_L) + (1-p)/u'(w_H)$ from our results above, we obtain

$$\left[\frac{p}{u'(w_L)} + \frac{1-p}{u'(w_H)} \right] \theta_H = \frac{px'(e_L)}{c'(e_L)} + \frac{(1-p)x'(e_H)}{c'(e_H)}$$

Moreover $\lambda_3 > \lambda_2$, since otherwise the first FOC, $(\lambda_3 - \lambda_2)u'(w_L) = p$, could not hold. Therefore $\lambda_3 > 0$, which means that its associated constraint, IC_L , binds. In particular, IC_L can be rewritten as

$$u(w_L) - \theta_L c(e_L) = u(w_H) - \theta_L c(e_H).$$

Then rearranging the right-hand side gives

$$u(w_L) - \theta_L c(e_L) = u(w_H) - \theta_H c(e_H) + (\theta_H - \theta_L) c(e_H),$$

and since PC_H binds, $u(w_H) - \theta_H c(e_H) = 0$, the first two terms on the right-hand side of the expression cancel, yielding

$$u(w_L) - \theta_L c(e_L) = (\theta_H - \theta_L) c(e_H),$$

which entails that the most efficient agent, θ_L , obtains in equilibrium a positive utility level, $(\theta_H - \theta_L)c(e)$, that increases in his difference with respect to the least efficient worker, $\theta_H - \theta_L$.

In contrast, the incentive compatibility condition of the least efficient worker IC_H , does not bind, implying that its associated Lagrange multiplier must be zero, $\lambda_2 = 0$. Using this result in the first and third FOCs yields

$$\lambda_3 = \frac{p}{u'(w_L)} \quad \text{and} \quad \frac{px'(e_L)}{c'(e_L)} = \lambda_3 \theta_L,$$

or, solving for λ_3 and combining the two FOCs,

$$\frac{p}{u'(w_L)} = \frac{px'(e_L)}{\theta_L c'(e_L)},$$

which means that, solving for $x'(e_L)$, we obtain

$$x'(e_L) = \frac{\theta_L c'(e_L)}{u'(w_L)}.$$

Hence, for the most efficient worker, the equilibrium outcome under asymmetric information coincides with the socially optimal result under symmetric information;

indeed the effort is increased until the point at which the MRS of the worker coincides with that of the firm. This result is common in screening problems, whereby the most productive agent suffers no distortion in his effort or output (i.e., “no distortion at the top”), and all other agents see their actions distorted relative to symmetric information.

Finally, using $\lambda_1 = p/u'(w_L) + (1-p)/u'(w_H)$, $\lambda_2 = 0$, and $\lambda_3 = p/u'(w_L)$ in the fourth FOC, we have

$$(1-p)x'(e_H) - \left[\frac{p}{u'(w_L)} + \frac{1-p}{u'(w_H)} \right] \theta_H c'(e_H) + \frac{p}{u'(w_L)} \theta_L c'(e_H) = 0,$$

and after rearranging, we obtain

$$\frac{(\theta_H - \theta_L)p}{1-p} \frac{c'(e_H)}{u'(w_L)} + \frac{\theta_H c'(e_H)}{u'(w_H)} = x'(e_H).$$

The effort level that solves this equation is hence the optimal effort under asymmetric information, e_H^{AI} . We can compare e_H^{AI} against the effort arising under symmetric information e_H^{SI} which, as shown in previous sections, solves $\theta_H c'(e_H)/u'(w_H) = x'(e_H)$. In particular, the first term in the equation is positive since $\theta_H - \theta_L > 0$, $p \in (0, 1)$, $c'(e_H) \geq 0$ and $u'(w_L) > 0$, implying that

$$\frac{(\theta_H - \theta_L)p c'(e_H)}{(1-p)u'(w_L)} + \frac{\theta_H c'(e_H)}{u'(w_H)} > \frac{\theta_H c'(e_H)}{u'(w_H)}.$$

Hence the effort level under asymmetric information is lower than that under symmetric information, $e_H^{AI} < e_H^{SI}$, as depicted in figure 10.11. Specifically, the left panel reproduces the FOC for optimal effort under symmetric information, $\theta_H c'(e_H)/u'(w_H) = x'(e_H)$, while the right panel illustrates the FOC under asymmetric information found above.

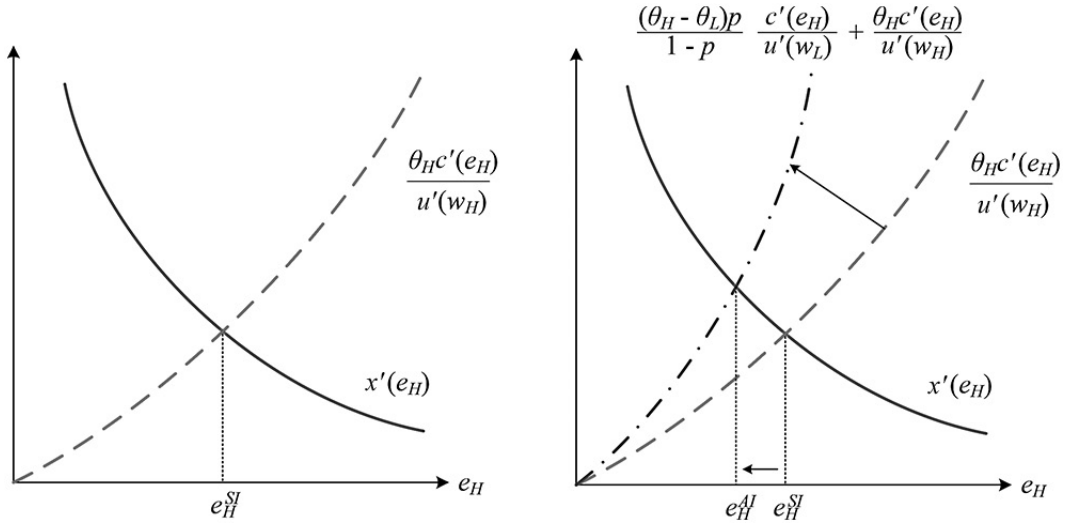
In summary, the pair of contracts (w_L, e_L) and (w_H, e_H) must satisfy the equations we found

$$u(w_L) - \theta_L c(e_L) = (\theta_H - \theta_L) c(e_H),$$

$$u(w_H) - \theta_H c(e_H) = 0,$$

$$x'(e_L) = \frac{\theta_L c'(e_L)}{u'(w_L)},$$

$$\frac{(\theta_H - \theta_L)p}{1-p} \frac{c'(e_H)}{u'(w_L)} + \frac{\theta_H c'(e_H)}{u'(w_H)} = x'(e_H).$$

**Figure 10.11**

Principal-agent problem under asymmetric information

Monotonicity in Effort

Another property to consider is that effort levels satisfy $e_L \geq e_H$, that is, the worker with the lowest cost of effort exerts a larger effort level than the worker with a high cost of effort. In order to show this property, let us combine IC_L and IC_H to obtain

$$u(w_L) - c(e_L, \theta_L) \geq u(w_H) - c(e_H, \theta_L) > u(w_H) - c(e_H, \theta_H) \geq u(w_L) - c(e_L, \theta_H),$$

where the first inequality originates from IC_L , the second inequality is due to the cost of effort satisfying $c(e_H, \theta_H) < c(e_H, \theta_L)$, and the third inequality is due to IC_H . Hence the triple inequality above can be rearranged as

$$c(e_H, \theta_L) - c(e_L, \theta_L) \geq u(w_H) - u(w_L) > c(e_H, \theta_H) - c(e_L, \theta_H).$$

Multiplying this expression by -1 , and using the first and last terms, yields

$$c(e_L, \theta_L) - c(e_H, \theta_L) < c(e_L, \theta_H) - c(e_H, \theta_H).$$

This condition indicates that the marginal cost of increasing effort from e_H to e_L is higher for the high-type than for the low-type worker. Therefore the worker with the low (absolute and marginal) cost of effort exerts a larger effort level than the high-type

worker.⁹ We can evaluate this condition in the cost of effort function $c(e, \theta) = \theta c(e)$, which yields

$$\theta_L [c(e_L) - c(e_H)] < \theta_H [c(e_L) - c(e_H)].$$

Since $\theta_L < \theta_H$ by definition, we must have that $c(e_L) > c(e_H)$; otherwise, the sign of the above inequality would be switched. Therefore, since the cost of effort function increases in e , $c(e_L) > c(e_H)$, effort is larger for the worker with the low cost of effort, $e_L > e_H$. ■

Example 10.7: Principal–agent problem under asymmetric information Let us now use the same functional forms as in the principal–agent model of example 10.6 to calculate the optimal contracts under asymmetric information. Taking the list of FOCs from above, we have

$$u(w_L) - \theta_L c(e_L) = (\theta_H - \theta_L) c(e_H) \Rightarrow w_L - e_L^2 = e_H^2,$$

$$u(w_H) - \theta_H c(e_H) = 0 \Rightarrow w_H = 2e_H^2,$$

$$x'(e_L) = \frac{\theta_L c'(e_L)}{u'(w_L)} \Rightarrow \frac{1}{e_L} = \frac{2e_L}{1} \rightarrow 2e_L^2 = 1 \rightarrow e_L = \frac{1}{\sqrt{2}} \cong 0.707,$$

$$\frac{(\theta_H - \theta_L)p}{1-p} \frac{c'(e_H)}{u'(w_L)} + \frac{\theta_H c'(e_H)}{u'(w_H)} = x'(e_H) \Rightarrow \frac{2e_H}{1} + \frac{4e_H}{1} = \frac{1}{e_H} \rightarrow e_H = \frac{1}{\sqrt{6}} \cong 0.408.$$

From the last two FOCs, we obtain the equilibrium effort levels $e_H = 1/\sqrt{6}$ and $e_L = 1/\sqrt{2}$. This leaves us with only two unknowns (the two salaries). In particular, from the first equation, $w_L - 1/2 = 1/6$, or $w_L = 2/3$; and from the second equation $w_H = 2/6 = 1/3$. Therefore, the optimal pair of contracts is

$$(w_H^{AI}, e_H^{AI}) = \left(\frac{1}{3}, \frac{1}{\sqrt{6}} \right) \approx (0.333, 0.408),$$

$$(w_L^{AI}, e_L^{AI}) = \left(\frac{2}{3}, \frac{1}{\sqrt{2}} \right) \approx (0.667, 0.707).$$

⁹ In a context with continuous effort, this condition can be expressed as $\partial c(e, \theta_L) / \partial e < \partial c(e, \theta_H) / \partial e$, and it is commonly known as the “single-crossing property” since it guarantees that the indifference curve of a θ_L -type worker and a θ_H -type worker cross each other only once.

From example 10.6, we can see that the introduction of asymmetric information entails no changes in effort for the worker with low cost of effort, $e_L^{SI} = e_L^{AI} = 0.707$ (i.e., “no distortion at the top” for the most efficient worker), but a lower effort for the high-cost worker than under symmetric information, $e_H^{AI} = 0.408 < 0.5 = e_H^{SI}$. Salaries are, in contrast, higher for the low-cost worker under asymmetric than symmetric information, namely $w_L^{AI} = 0.667 > 0.5 = w_L^{SI}$; but lower for the high-cost worker, namely $w_H^{AI} = 0.333 < 0.5 = w_H^{SI}$. Overall, the net utility that each type of worker obtains under asymmetric information is

$$u_H^{AI} = w_H - 2e_H^2 = 0,$$

$$u_L^{AI} = w_L - e_L^2 = \frac{1}{6} \cong 0.167,$$

which entails that the worker with a low cost of effort captures an information rent of $u_L^{AI} - u_L^{SI} = 0.167 - 0 = 0.167 > 0$, while the worker with a high cost of effort does not, $u_H^{AI} = u_H^{SI} = 0$. Intuitively, the firm must compensate the low-cost worker above symmetric information terms in order for him to reveal his type (i.e., to choose the contract meant for him), while the worker with a high cost of effort is not compensated above his symmetric information utility level. Additionally we can compare the effort levels and wages between both information contexts as follows:

$$e_L^{AI} = 0.707 = 0.707 = e_L^{SI},$$

$$e_H^{AI} = 0.408 < 0.500 = e_H^{SI},$$

$$w_L^{AI} = 0.667 > 0.500 = w_L^{SI},$$

$$w_H^{AI} = 0.333 < 0.500 = w_H^{SI}.$$

Extensions The principal–agent problem considered above can be extended in many dimensions. We next provide some examples commonly examined in the literature: (1) allowing for several principals to compete for the labor services of one common agent (as described in Stole 1990 and Martimort 1996); (2) allowing for the agent’s privately observed type θ to be multidimensional (i.e., a vector), such as his productivity in different tasks (see Laffont et al. 1987; Rochet 1996); (3) allowing for the principal to assign different tasks to the agent (where at least one of the tasks can be done by either the principal or the agent) in order to identify conditions under which the principal profitably delegates the task, or instead prefers to carry it out himself (as in Riordan

and Sappington 1987); (4) considering a principal–agent relationship that lasts for more than one period (i.e., repeated adverse selection), and how the agent’s incentives to conceal his type from the principal can increase with and without commitment (see Baron and Besanko 1984; Laffont and Tirole 1988, 1990); (5) allowing the principal to incorrectly observe realized profits (as analyzed by Caillaud et al. 1992); or (6) permitting the contract between the principal and the agent to be dynamic, thus spanning several periods (see Bolton and Dewatripont 2005, chapter 9, and references therein).

10.6 Application of Adverse Selection—Regulation

There are several contexts in which regulatory agencies cannot observe some characteristic of the regulated firm or of individual. Examples include a firm’s production costs (in the case of setting production subsidies), its costs from pollution abatement (when regulators set emission fees or quotas seeking to curb pollution), and a consumer’s willingness to pay for certain products (e.g., when governments design tax rebates for the purchase of solar panels or home insulating materials). In these scenarios the privately informed party (e.g., firm) has the incentives to overstate its costs and, as a consequence, the regulator cannot directly ask firms about their production costs since responses would be unreliable. As we next show, the above-mentioned results in adverse selection models offer an alternative contracting tool for regulators to use in order to extract information from firms (or consumers) that are privately informed about a relevant characteristic, such as their production costs, that the government can use to design public policies.

Consider a government regulating a monopoly with cost function $c(q) = C + cq$, where C denotes its fixed costs and $c > 0$ its marginal costs. The consumer pays F for the bulk of q units consumed, and the monopolist may receive a lump-sum subsidy from the government of S . Assume that the shadow cost of raising public funds is $g \in (0, 1)$, thus implying that the total cost of providing a subsidy S to the monopolist is $(1 + g)S$.¹⁰ For completeness, we first analyze the first-best outcome, which arises when the government has symmetric information about the monopolist’s costs, and then we will move to different regulatory context in which the government does not accurately observe monopolist’s costs.

¹⁰ The shadow cost of raising public funds can be understood as the deadweight loss that arises from setting a sales tax (or a payroll tax). Recall from previous chapters that a deadweight loss arises from these taxes unless individuals and firms are completely unresponsive to the tax (inelastic demand and supply).